

EPFL Computer Science student (BSc, expected 2027) building machine learning systems and quantitative trading research from first principles — from neural networks implemented in raw NumPy to a full look-ahead-safe backtesting platform tested on real market data. Seeking a quantitative research internship.

EDUCATION

École Polytechnique Fédérale de Lausanne

BSc in Computer Science · expected 2027

Coursework: Machine Learning, Probability & Statistics, Algebra, Analysis, Algorithms & Data Structures, Software Construction, Computer Architecture

Lycée International de Saint-Genis-Pouilly

French Baccalauréat, High Honors · 2024

TECHNICAL SKILLS

Programming

Python, Java, C, Scala, SQL, Verilog, RISC-V Assembly, JavaScript

Quantitative

Pandas, NumPy, backtesting, walk-forward validation, risk analytics (Sharpe, Sortino, VaR/CVaR), bootstrap & permutation testing, transaction-cost modeling

Machine Learning

Neural networks (forward/backprop in NumPy), K-Means, logistic & linear regression, gradient descent, L2 regularization, hyperparameter tuning, model evaluation

CERTIFICATIONS

CS50x — Introduction to Computer Science

Harvard University, 2025

CS50P — Programming with Python

Harvard University, 2025

LANGUAGES

French (native) · English (B2) · Turkish (B2)

VOLUNTEERING

Food bank collections, since 2022

Robotics workshops with PanglossLabs (FabLab) and Centre SocioCulturel Les Libellules

INTERESTS

Quantitative finance · Machine learning · Tennis · Mountain hiking

QUANTITATIVE & TECHNICAL PROJECTS

QuantLab — Reproducible Quantitative Research Platform

2026

Personal project · github.com/sefaav/QuantLab · Python, Pandas, NumPy

- Modular platform turning a hypothesis into a reproducible, bias-aware experiment: data validation, signals, allocation, cost-aware execution, risk metrics and reporting, driven by one YAML config.
- Look-ahead-safe engine: signal at t , position at $t+1$, realised return — enforced and unit-tested; commission, spread and volume-based slippage modeled, gross vs net always reported.
- Strategy library behind one interface — momentum (time-series and cross-sectional), mean reversion, trend following, pairs trading (hedge ratio, ADF test) on Yahoo Finance and Binance data.
- Risk analytics from first principles (Sharpe, Sortino, Calmar, drawdown, VaR/CVaR, alpha/beta) and a robustness suite: walk-forward with parameter selection on validation only, sensitivity heatmaps, bootstrap and permutation tests.
- Reports every result as computed, losing strategies included, with an automatic limitations section and five executed notebooks; >80% core test coverage, Ruff and mypy clean, CI, pinned lockfile.

Machine Learning from Scratch — MLP, K-Means & Model Selection

2026

EPFL coursework (CS-233) · Python, NumPy, Matplotlib

- Implemented a multi-layer perceptron in raw NumPy — fully connected layers, forward pass, backpropagation, mini-batch gradient descent — with softmax cross-entropy for classification and an identity output with MSE loss for regression.
- Added L2 regularization, validation-loss monitoring and early stopping with best-weight restoration; implemented K-Means with vectorized distances, empty-cluster handling and majority-vote labelling.
- Designed a validation-based grid search over architecture, learning rate, batch size and regularization, with an evaluation pipeline comparing accuracy, macro F1, MSE and confusion matrices.
- On a behavioural dataset, the best MLP reached 87.0% test accuracy and 0.811 macro F1 (MSE 0.999 on the regression target), outperforming K-Means; cross-entropy improved recall on the rare minority class.

Multicycle RISC-V CPU

2025

EPFL coursework (CS-200) · Verilog

- Designed a 32-bit multicycle RISC-V CPU: datapath, control FSM, interrupt handling, memory-mapped I/O; verified with Verilator simulation and GTKWave waveform debugging.

Additional projects

ReCHor (Java) — Pareto-optimal public-transport route planner, offline multi-criteria search. **EPFL Events** (Scala) — real-time multi-user event platform, concurrency-safe state-machine backend.

EXPERIENCE

Bäckerei & Konditorei Tschorek — Vienna, Austria

Jul–Aug 2024

Seasonal bakery role: cashier duties and production support.

“Chez Bob” — Saint-Palais-sur-Mer, France

Jun–Jul 2022

Kitchen prep in a high-volume seaside restaurant, up to 1,500 covers per service, 20-person team.

Sporticamp & Tennis School — Saint-Genis-Pouilly, France

Sep 2022–Jun 2023

Assistant tennis coach, children aged 4–12.

Aicogy (Coding & Robotics) — Ferney-Voltaire, France

2017–2020

Teaching assistant for coding and robotics workshops, ages 6–15.